1..DTFT of x[n] = cos((2pi n)/3) + sin((2pi n)/5)

```
clc;
clear;
close all;

%% Q1: DTFT using formula

n = 0:99;
x = cos(2*pi*n/3) + sin(2*pi*n/5);

w = linspace(-2*pi,2*pi,2000);

X = zeros(size(w));

for k = 1:length(w)
    X(k) = sum(x .* exp(-1j*w(k)*n));
end

figure;

subplot(2,1,1);
plot(w/pi,abs(X),'LineWidth',1.5);
grid on;
xlabel('\omega/\pi');
ylabel('|X(e^{j\omega})|');
title('DTFT Magnitude');

subplot(2,1,2);
plot(w/pi,angle(X),'LineWidth',1.5);
grid on;
xlabel('\omega/\pi');
ylabel('Phase (rad)');
title('DTFT Phase');

%% Q1(b): Using freqz

figure;
freqz(x,1,2000);
```

2…symmetric rectangular pulse is given by N=2,5,10,15

```
clc;
```

```
clear;
close all;

%% Q2: Rectangular pulse

N_values = [2 5 10 15];
w = linspace(-pi,pi,2000);

figure;

for i = 1:length(N_values)

    N = N_values(i);

    n = 0:N-1;
    x = ones(1,N);

    X = zeros(size(w));

    for k = 1:length(w)
        X(k) = sum(x .* exp(-1j*w(k)*n));
    end

    % Normalize so X(e^j0) = 1
    X_norm = X/N;

    subplot(2,2,i);
    plot(w/pi,abs(X_norm),'LineWidth',1.5);
    grid on;
    xlabel('\omega/\pi');
    ylabel('|X(e^{j\omega})|');
    title(['N = ',num2str(N)]);

end

sgtitle('Normalized DTFT of Rectangular Pulse');
```

3…DTFT Determine the DTFT of the sequence y[n]= x[n]*x[n], for N = 2 ,5, 10, 15.?

```
clc;
clear;
close all;
```

```
%% Q3: y[n] = x[n] * x[n]

N_values = [2 5 10 15];
w = linspace(-pi,pi,2000);

figure;

for i = 1:length(N_values)

    N = N_values(i);

    % Rectangular pulse
    x = ones(1,N);

    % Convolution
    y = conv(x,x);

    n = 0:length(y)-1;

    Y = zeros(size(w));

    for k = 1:length(w)
        Y(k) = sum(y .* exp(-1j*w(k)*n));
    end

    % Normalize so Y(e^j0) = 1
    Y_norm = Y / sum(y);

    subplot(2,2,i);
    plot(w/pi,abs(Y_norm),'LineWidth',1.5);
    grid on;
    xlabel('\omega/\pi');
    ylabel('|Y(e^{j\omega})|');
    title(['N = ',num2str(N)]);

end

sgtitle('Normalized DTFT of y[n] = x[n] * x[n]');
```

4... Determine H(z), H(e) and plots its pole-zero diagram magnitude and phase responses using MATALB. Identify the nature of the system by observing the frequency response of the system.

i) y[n] = 1/3 * [x[n] + x[n - 1] + x[n - 2]]

ii) y[n] = x[n] - x[n - 8]

```
clc;
clear;
close all;

%% Q4(i)
% y[n] = 1/3 [x[n] + x[n-1] + x[n-2]]

b1 = [1/3 1/3 1/3];
a1 = 1;

figure;
freqz(b1,a1,2000);
sgtitle('System (i): H(z) = 1/3(1+z^{-1}+z^{-2})');

figure;
zplane(b1,a1);
title('Pole-Zero Plot - System (i)');


%% Q4(ii)
% y[n] = x[n] - x[n-8]

b2 = [1 zeros(1,7) -1];
a2 = 1;

figure;
freqz(b2,a2,2000);
sgtitle('System (ii): H(z) = 1-z^{-8}');

figure;
zplane(b2,a2);
title('Pole-Zero Plot - System (ii)');


%% Q4(iii)
% y[n] = 1/2 x[n] + 1/2 x[n-1]

b3 = [1/2 1/2];
a3 = 1;

figure;
```

```
freqz(b3,a3,2000);
sgtitle('System (iii): H(z) = 1/2(1+z^{-1})');

figure;
zplane(b3,a3);
title('Pole-Zero Plot - System (iii)');
```

…DTSP??
1….x(t) = cos (2 π (100) 1) +2 cos 920 nt, sampled at

i. 1280 samples/sec (Understanding frequency resolution)

ii. 2560 samples/sec (Understanding spectral leakage due to non resolvable Components) Consider N = 64

```
clc;
clear;
close all;

N = 64;

%% Sampling frequency = 1280 Hz
Fs = 1280;
n = 0:N-1;
t = n/Fs;

x = cos(2*pi*100*t) + 2*cos(2*pi*460*t);

X = fft(x,N);
f = (-N/2:N/2-1)*(Fs/N);

X_mag = abs(fftshift(X))/N;

figure;
plot(f,X_mag,'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Q1: Spectrum, Fs = 1280 Hz');

%% Sampling frequency = 2560 Hz
Fs = 2560;
```

```
t = n/Fs;

x = cos(2*pi*100*t) + 2*cos(2*pi*460*t);

X = fft(x,N);
f = (-N/2:N/2-1)*(Fs/N);

X_mag = abs(fftshift(X))/N;

figure;
plot(f,X_mag,'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Q1: Spectrum, Fs = 2560 Hz');
```

2…Spectrum leakage due to zero padding(n) =cos 27 7,0≤n≤50, choose N = 64

```
clc;
clear;
close all;

Fs = 1280;
N = 64;

n = 0:50;

% Original signal
x = cos(2*pi*(80/1280)*n);

% Zero padding to N = 64
x_padded = [x zeros(1,N-length(x))];

X = fft(x_padded,N);

f = (-N/2:N/2-1)*(Fs/N);

X_mag = abs(fftshift(X))/length(x);

figure;
plot(f,X_mag,'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
title('Q2: Spectrum with Zero Padding');
```

3…Understanding aliasing. x(t) = cos(2pi(700) * t) sampled at 1280 Hz and choose N = 64

```
clc;
clear;
close all;

Fs = 1280;
N = 64;

n = 0:N-1;
t = n/Fs;

% Original signal: 700 Hz
x = cos(2*pi*700*t);

X = fft(x,N);

f = (-N/2:N/2-1)*(Fs/N);

X_mag = abs(fftshift(X))/N;

figure;
plot(f,X_mag,'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Q3: Aliasing Spectrum');
```

4…It is desired to use the IFFT as a waveform generator to synthesize a waveform described by the following discrete function.
chi(nT) = cos(2pi*fnT) where sampling frequency f_{s} = 64Hz N = 64 point (FFT Size) f = 3Hz

```
clc;
clear;
close all;

%% Q5: IFFT waveform generation
```

```
Fs = 64;
N = 64;
f0 = 3;

% Frequency-domain signal
X = zeros(1,N);

% Frequency bin corresponding to 3 Hz
k = f0*N/Fs;

X(k+1) = N/2;
X(N-k+1) = N/2;

% IFFT
x = ifft(X);

n = 0:N-1;
t = n/Fs;

figure;
stem(n,x,'filled');
grid on;
xlabel('n');
ylabel('x[n]');
title('Q5: Waveform Generated using IFFT');

figure;
plot(t,x,'LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('x(t)');
title('Q5: Reconstructed Cosine Wave');
```

Exp5. Generation of Random Signals and their Distributions

1…Generate random sequences and plot histograms?

```
clc;
clear;
close all;

N = 10000;
```

```
%% a) Uniform distribution (0,1)
x1 = rand(1,N);

figure;
histogram(x1,50);
grid on;
xlabel('Value');
ylabel('Frequency');
title('Uniform Distribution (0,1)');


%% b) Gaussian distribution
% Mean = 0, Variance = 1
x2 = randn(1,N);

figure;
histogram(x2,50);
grid on;
xlabel('Value');
ylabel('Frequency');
title('Gaussian Distribution: Mean = 0, Variance = 1');


%% c) Gaussian distribution
% Mean = 2, Variance = 4
% Standard deviation = sqrt(4) = 2

x3 = 2 + 2*randn(1,N);

figure;
histogram(x3,50);
grid on;
xlabel('Value');
ylabel('Frequency');
title('Gaussian Distribution: Mean = 2, Variance = 4');


%% d) Uniform integer distribution (1 to 5)

x4 = randi([1 5],1,N);

figure;
histogram(x4,'BinMethod','integers');
grid on;
```

```
xlabel('Integer Value');
ylabel('Frequency');
title('Uniform Integer Distribution (1 to 5)');
```

2…Generate complex random sequences of length 10000, real and imaginary parts of the sequence Gaussian distributed with zero mean and variance two. Plot the histogram of the magnitude and phase of the complex random sequence

```
clc;
clear;
close all;

N = 10000;

%% Complex Gaussian random sequence

real_part = sqrt(2)*randn(1,N);
imag_part = sqrt(2)*randn(1,N);

z = real_part + 1j*imag_part;

%% Magnitude
mag = abs(z);

figure;
histogram(mag,50);
grid on;
xlabel('Magnitude');
ylabel('Frequency');
title('Histogram of Magnitude of Complex Gaussian Signal');


%% Phase
phase = angle(z);

figure;
histogram(phase,50);
grid on;
xlabel('Phase (radians)');
ylabel('Frequency');
title('Histogram of Phase of Complex Gaussian Signal');
```

3…Check the validity of the central limit theorem using MATLAB command.

```
clc;
clear;
close all;

N = 10000;

%% Central Limit Theorem

M = [1 2 5 10 20 50];

figure;

for i = 1:length(M)

    m = M(i);

    % Generate uniform random variables
    x = rand(N,m);

    % Sum and normalize
    y = sum(x,2);

    % Normalize to zero mean and unit variance
    y = (y - mean(y))/std(y);

    subplot(2,3,i);
    histogram(y,40);
    grid on;
    title(['M = ',num2str(m)]);
    xlabel('Value');
    ylabel('Frequency');

end

sgtitle('Central Limit Theorem');
```

4…Gaussian random sequence through square-law device

```
clc;
clear;
close all;

N = 10000;
```

```
%% Gaussian random sequence
% Mean = 0
% Variance = 1

X = randn(1,N);

%% Square-law device
Y = X.^2;

%% Plot histogram

figure;
histogram(Y,50);
grid on;
xlabel('Y = X^2');
ylabel('Frequency');
title('Histogram of Output of Square-Law Device');
```